Internal and international migration of scholars worldwide

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Introduction

- Theoretical impressions
  - Internal versus international migration (*1*, *2*)

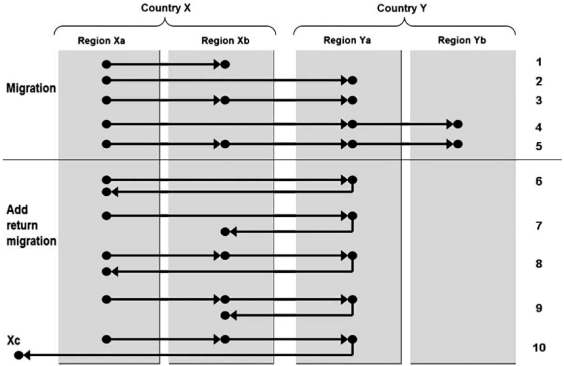

Figure 1. Migration pathways

(King, Skeldon (2010))

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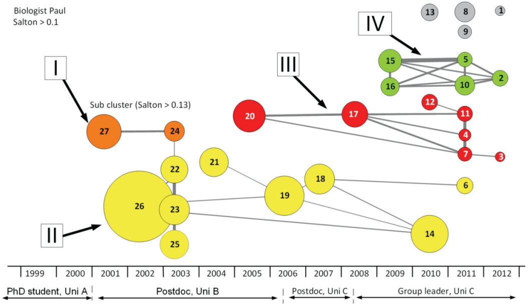
Figure 4: Cognitive Career of Biologist Paul



(Laudel, Glaser (2015))

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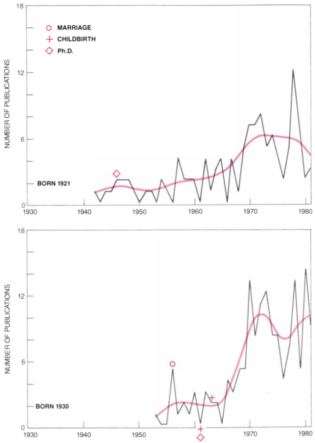
- Methodological impressions
  - Using bibliometric data alongside demographic life events (*4*)



LOWER RATES OF PUBLICATION in the early part of a career are characteristic of both married men and single women. The publication profile of a distinguished woman biologist (*top*) who never married shows the same pattern of oscillations and an overall increase as the graphs of women who married and had children. The same pattern can be seen in the profile of an eminent male chemist (*bottom*). He published at a much slower pace when his children were young, although his domestic responsibilities were minimal.

(Cole, Zuckerman (1987))

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  - Repurposing bibliometric data for knowledge transfer (*5*, *6*)

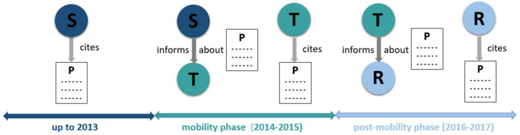

Figure 3. Illustration of the knowledge transfer process from source of knowledge S to the transmitter T who is the mobile scientist. The recipient R of the knowledge is a coauthor of T and receives the knowledge of the publication P upon return of T to Germany.

(Aman (2020))

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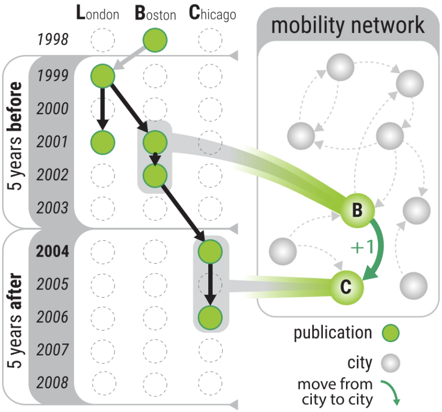
  - Repurposing bibliometric data as network of scientific mobility among cities (*7*)



(Verginer and Riccaboni (2020))

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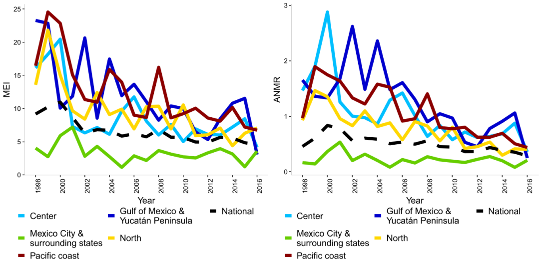
  - Repurposing bibliometric data for migration research (*8–11*)



(b) Migration Effectiveness Index

(c) Aggregate Net Migration Rate

Figure 8 Migration intensity indices, over time. Dashed line represent country indices

(Miranda Gonzalez et al. (2020))

Our proposal

Repurposing bibliometric data for:

Internal and **international** migration of **scholars** worldwide

A simple and scalable idea:

Changes in **institutional affiliations** can be used to infer changes in **residence over time** for individual scholars and for populations

Our questions

1. Which are the (persistently) **attractive** sending/receiving regions internally/internationally?
2. Is academic **talent circulation** happening more *within* or *between* countries?
3. What are the **mutual effects** of these two systems of migration?

Data and measures

- **Scopus 1996-2020** article/review publications (**28+** million, **8+** million scholars)
- Provided by German Competence Network for Bibliometrics via Max Planck Digital Library
- Encoding **country** of affiliation from bibliometric record (not perfect, need proper disambiguation)
- **Organization name disambiguation** using ROR API (based on *12*), GeoNames codes for intra-country regions
- **Author name disambiguation** based on Scopus author IDs (**98.3%** precision (no publication by others) and **90.6%** recall (all publications by X)) (*13*)

$$NMR_{i,t,k} = 1000 \times \frac{I_{i,t,k} - E_{i,t,k}}{N_{i,t}} \tag{1}$$

$$ANMR_t = 100 \times \frac{0.5 \sum_i |I_{i,t,k} - E_{i,t,k}|}{\sum_i N_{i,t}} \tag{2}$$

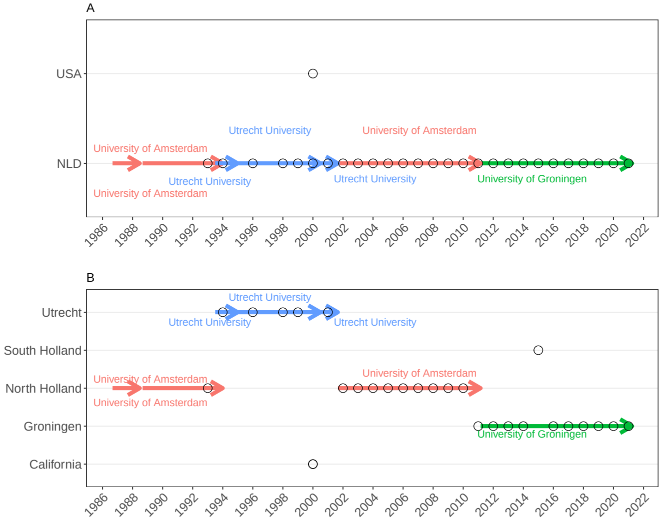
$$MEI_t = 100 \times \frac{\sum_i |I_{i,t,k} - E_{i,t,k}|}{\sum_i (I_{i,t,k} + E_{i,t,k})} \tag{3}$$

i: sub-national region. t: year. $N_{i,t}$: total number of scholars. Subscript k: type of data, internal or international. $I_{i,t}$ inflow of scholars. $E_{i,t}$ outflow.

$$log\ counts \sim s(year) + s_{fs}(year, region) + offset$$

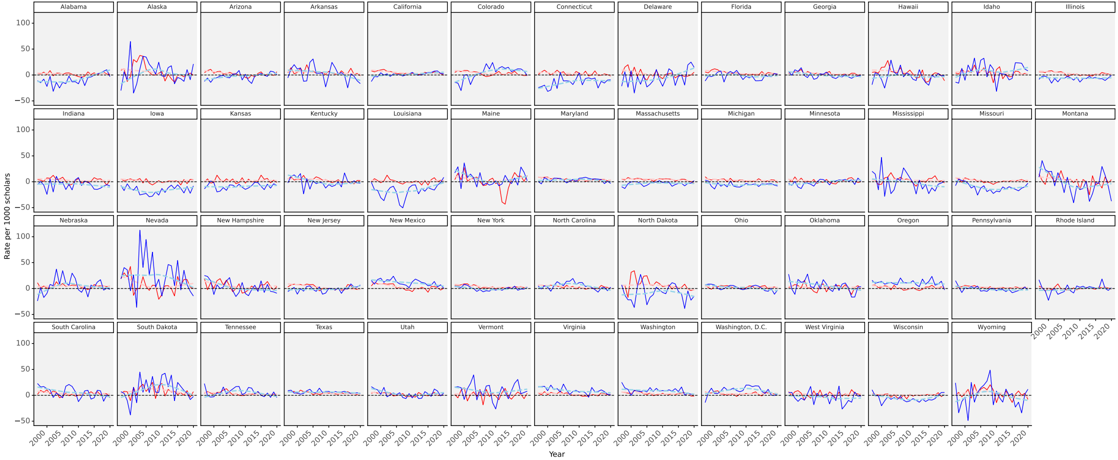
$$log\ counts \sim s(year) + s_{re}(year, region) + s_{re}(region) + offset$$

Example of integrated internal/international framework



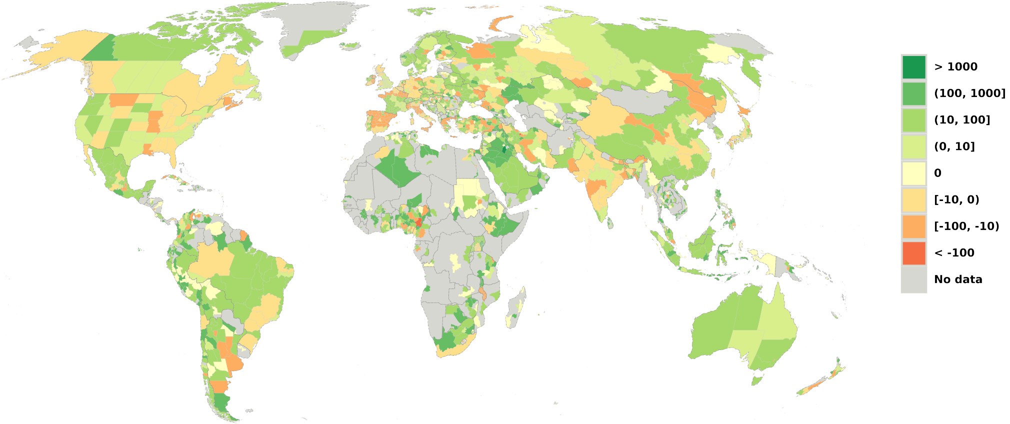
USA, Region's NMR, international (red), internal (blue)

Internal senders: Alabama, Connecticut, Delaware, Iowa, Kansas, Louisiana and Missouri; **Internal receivers**: Colorado, Nevada, New Mexico, Oregon, South Dakota, Virginia, Washington and Washington, D.C.; **International receivers**: Alaska, North Dakota and South Dakota

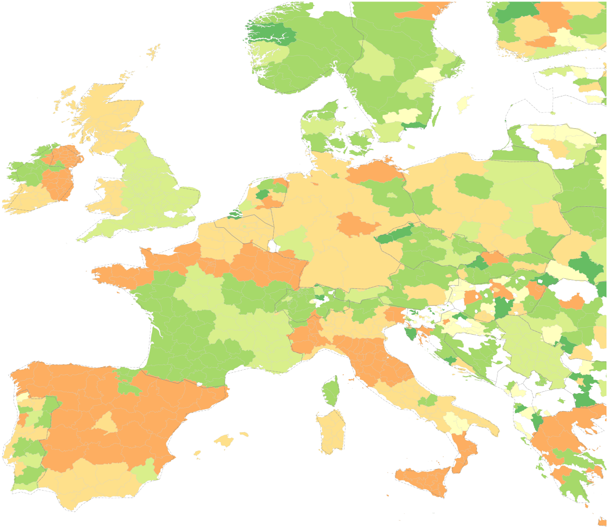


World

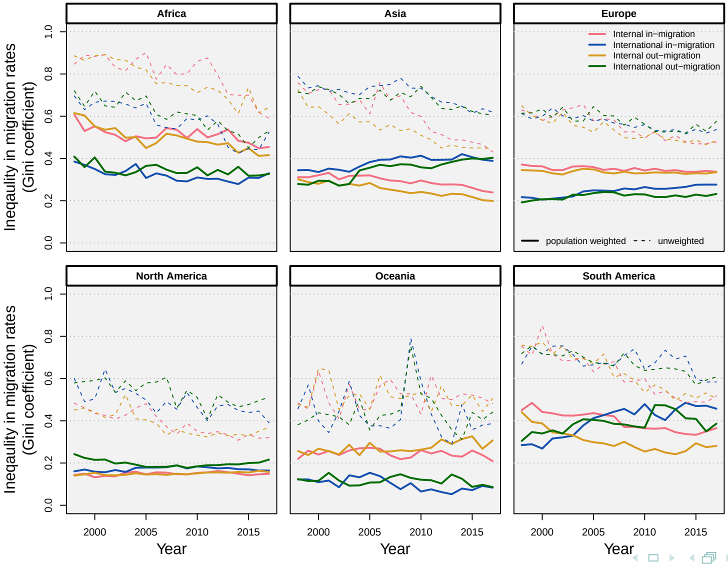
Internal and international NMR (per 1000 scholars), 2012-17



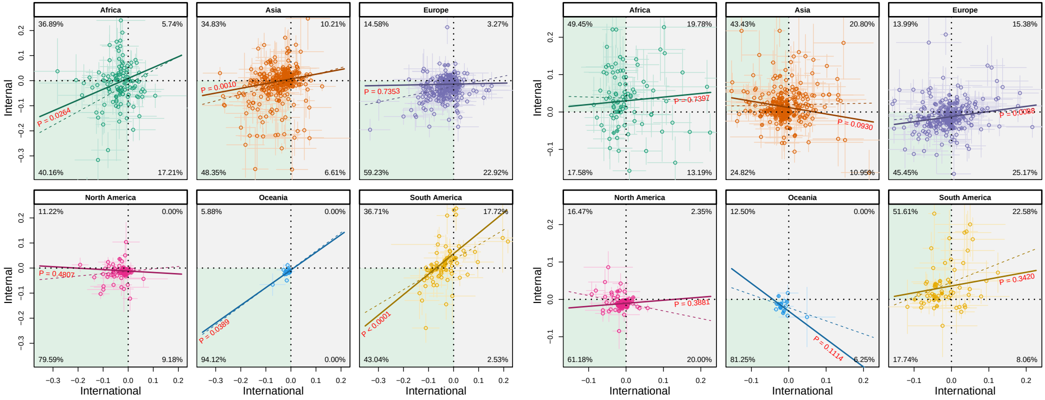
Europe



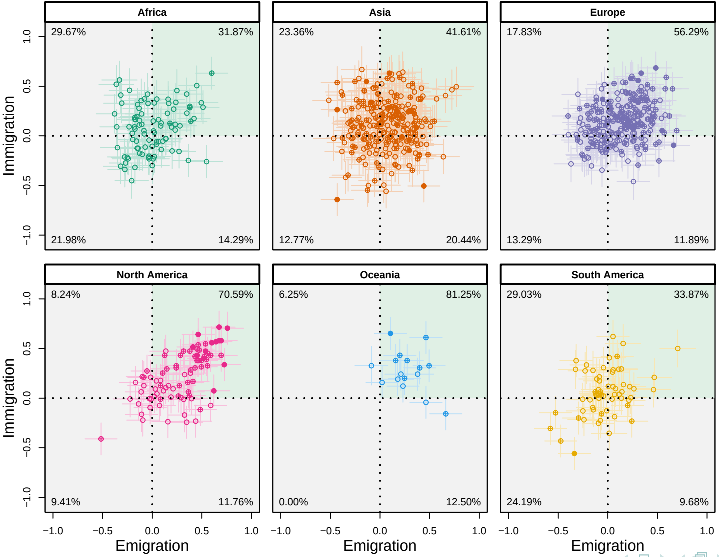
Heterogeneity (and size dependency) in migration rates



In-migration (left), Out-migration (right)



Inter-relatedness of in-migration and out-migration



Limitations

- Scopus published scholars (who publish actively and multiple times)
- Scopus coverage issues (over-representation of WEIRD countries, English language, hard sciences, etc.)
- Author's academic birth is not equivalent to their nationality and some of the outmigrations could be **return** migration.
- and many other limitations ...

Discussion and conclusions

- Our methodology in re-purposing bibliometric data offers unprecedented opportunities to study ***internal*** and ***international*** scholarly mobility
- This is the first **global database of migration of scholars at internal/international levels** (8% mobile international versus 12.4% internal).
- On a global scale, we developed **migration indexes**, NMR, ANMR, and MEI.
- **A clear pattern is found in inmigration** rates in all continents where an **attractive region**, receiving scholars from international origins is also attractive in receiving scholars from internal origins.
- **The reverse**, being a popular sending origin to internal and international destinations is not always true.
- **A clear pattern is found in inter-relatedness** where both in- and out-migration over time increase or decrease in agreement.

Extensions and next steps

- Word2vec word embedding for multiple affiliations
- Identifying nationality using author's first and last name and trajectory of all affiliations to go closer to nationality (or families of geographical families, e.g., Spanish names, Asian names, ...)
- Comparison with employment history in ORCID as (hopefully) less biased towards WEIRD countries
- Gender, discipline, ..., differences

Thank you!

Questions and comments are welcome!

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or
`https://www.demogr.mpg.de/papers/working/wp-2023-038.pdf`

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